



D2 · DELIVERABLE

Software Requirements Specification

Explainability, configurable criteria, audit logging and the fairness threshold, as measurable and traceable requirements.

IEEE 830-style SRS for HireSense. Every requirement carries an acceptance criterion and traces to the module that implements it and the test that verifies it.

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements for **HireSense**, the rebuilt applicant-screening system of a staffing company that screens **90,000 applications a year**. It follows the structure of IEEE Std 830-1998. It is written so that every requirement is *testable*: each one has a measurable acceptance criterion and traces to a design element and a test (§7).

1.2 Scope

HireSense ingests PDF résumés, parses them with a confidence score, screens them with the specified routine against explicit per-job criteria, ranks candidates by explicit keys, logs every decision append-only, and runs a recurring bias audit with a defined consequence on breach. Out of scope: interview scheduling, offers, payroll, and candidate-facing job boards.

1.3 Problem statement (from the case study)

The current tool ranks candidates and recruiters trust the ranking. Nobody can explain why candidate 412 ranked above candidate 87. An internal review found the ranking correlated with the applicant's university tier far more strongly than with later job performance. Résumé parsing works on 71% of documents and fails silently on the rest.

1.4 Definitions

TERM	DEFINITION
Parse confidence	Weighted share of decision fields extracted reliably, × text-layer factor (1, or mean OCR word confidence). Range 0–1.
Criteria version	Immutable snapshot of a job's mandatory skills, bands, threshold, city and remote flag.
Selection rate (SR)	Shortlisted ÷ decided applications for a group in an audit window.
Impact ratio	SR of a group ÷ highest SR among comparable groups (4/5ths rule when ≥ 0.80).
Declared group	Group from voluntary self-identification (gender, age band), or derived for audit only (university tier).
Silent failure	A parse that is wrong but has confidence ≥ 0.6 , so it is decided automatically.
Shadow mode	The legacy system runs on the same input for comparison but never affects decisions.

1.5 References

- Case Study 31, SEPM, B.Tech CSE (given data used verbatim).
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- I. Sommerville, *Software Engineering* (prescribed text): requirements engineering and validation.
- Uniform Guidelines on Employee Selection Procedures, 29 CFR §1607.4(D): the four-fifths rule.
- Digital Personal Data Protection Act, 2023 (India).

2 Overall description

2.1 Product perspective

HireSense replaces the legacy screener without stopping screening: it runs in **shadow mode** beside the legacy system until each increment's exit criteria are met (Project Plan, D5). It is a web application (Next.js) with a screening API (Bun + Hono), a SQLite store, an optional on-device OCR model and an optional local language model.

2.2 Stakeholders and user classes

STAKEHOLDER	INTEREST	INFLUENCE	KEY NEEDS
Recruiter	Fast, defensible shortlists	High	Ranked list with reasons; trust that unreadable résumés are not dropped
Reviewer (talent ops)	Clear manual-review queue	Medium	See why an application was routed; correct fields quickly
Criteria owner / hiring manager	Hire the right people	High	Configure criteria without engineers; preview impact
Compliance / audit lead	Defensible, non-discriminatory process	High	Recurring audit, evidence, defined breach response
Head of Talent	Reputation, throughput	High	Escalations, KPIs
Candidates	Fair and explainable treatment	Low (individually) · high (collectively, reputational)	Not rejected because a parser failed; data protection
Engineering	Maintainable system	Medium	Isolation of parsing and ranking; tests; versioning

2.3 Product functions

- Ingest résumé PDFs; parse with confidence; OCR fallback for scans.
- Configure, version and preview ranking criteria per job.
- Screen with the specified routine; rank by explicit keys; explain any decision or ordering.
- Append-only decision log with search and CSV export.
- Manual review queue with optional AI-assisted extraction.
- Recurring bias audit with severity grading and automatic consequence.

2.4 Constraints

- Use the case-study figures as given: the routine, the 0.6 confidence gate, the 29% failure rate, and the 80% rule.
- Screening must continue during the rebuild.
- Self-declared attributes must never be inputs to screening.
- Runs on commodity hardware (reference: 8 GB laptop).

2.5 Assumptions and dependencies

- Self-identification is voluntary. A non-trivial share answers "Prefer not to say" (85 of 2,000 in the demo data).
- University tiers are an illustrative list used only to audit for the proxy the internal review found.
- All candidate data in the prototype is synthetic.

3 Requirement elicitation and analysis

Requirements were elicited by **document analysis** of the case study and its internal-review findings, by **scenario analysis** of the failure modes (silent parse failure, unexplainable ranking, tier correlation), and by **prototyping**: a working prototype was built and measured against ground truth, and it exposed requirements the text did not state, such as the anomaly cap and the minimum group size.

PROBLEM IN THE CASE STUDY	ROOT CAUSE	REQUIREMENT RESPONSE
Nobody can explain why 412 ranked above 87	Opaque score; implicit ordering	FR-S1-S3: specified routine, stored trace, explicit rank keys
Ranking tracked university tier	Tier used directly or by proxy	FR-S4 (excluded inputs) + FR-F1-F4 (audit catches proxies)
Parsing fails silently on 29%	No confidence; failure looks like 'no skills'	FR-P1-P4: confidence, routing, OCR, acceptance threshold
Rejections not traceable	No reason logging	FR-L1-L3: append-only log with reasons

4 Functional requirements

4.1 Résumé parsing

The case study states parsing works on 71% of documents and fails silently on the rest. These requirements make every parse result carry its own reliability, and make failure visible.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-P1 Parse with confidence	The system shall extract skills, years of experience, city, name and email from a PDF résumé and attach a parse confidence in [0, 1] with a per-field breakdown.	Every stored parse has confidence $\in [0,1]$ and a breakdown of 5 weighted fields summing to 1.	Must
FR-P2 No silent failure	A résumé with no usable text layer (< 200 characters) or with a recognised section that yields no data shall receive confidence < 0.6 and must never be rejected automatically.	0 automated rejections of documents with confidence < 0.6 (decision log query).	Must
FR-P3 OCR fallback	Where the PDF has no text layer, the system should recover text by OCR and scale confidence by the mean OCR word confidence.	Scanned-résumé parse success $\geq 95\%$ on the evaluation corpus.	Should
FR-P4 Parse acceptance threshold	Released parser versions shall correctly parse $\geq 95\%$ of the evaluation corpus with $\leq 1\%$ silent failures (wrong but confidence ≥ 0.6).	Measured by seed/evaluate.ts against ground truth on every build.	Must

4.2 Configurable ranking criteria

Ranking criteria must be explicit and configurable. Criteria are data, never code.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-C1 Explicit criteria per job	Each job shall have explicit criteria: mandatory skills, experience bands with scores, shortlist threshold, job city and a remote-allowed flag.	Criteria schema validated on save; invalid criteria rejected with a message.	Must
FR-C2 Versioned changes	Changing criteria shall create a new immutable version with author, timestamp and a mandatory change note; every decision shall reference the version it used.	100% of decision_log rows carry a criteria_version_id.	Must

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-C3 Impact preview	Before publishing, a recruiter should be able to dry-run draft criteria on all applicants of the job and see outcome shifts and the fairness check.	Preview returns outcome counts and impact ratios without writing to the log.	Should
FR-C4 Ranking change isolation	A change to a ranking rule shall not require any change to the parsing module.	ranking/ and scoring/ import nothing from parsing/ (static dependency check).	Must

4.3 Screening and explainability

Nobody could explain why candidate 412 ranked above candidate 87. Every decision and every rank order must be explainable from stored facts.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-S1 Specified routine	Screening shall implement the routine exactly as given: missing mandatory skills → reject; else score experience band; shortlist if score ≥ threshold and location matches, else if score ≥ threshold and remote allowed; else waitlist; route to manual review if parse confidence < 0.6.	All 8 CFG paths produce the specified outcome (D1).	Must
FR-S2 Decision explanation	Every decision shall store a reason code, a human-readable reason and the trace of CFG nodes visited with the values tested at each.	100% of decisions have non-empty reason and a trace ending at N9/N10.	Must
FR-S3 Explicit rank order	Candidates of a job shall be ordered by explicit keys only: outcome, then band score, then application time, then application id; the system shall answer "why is A above B" with the first differing key.	Compare view returns winner and key for any two applications.	Must
FR-S4 Protected attributes excluded	Screening and ranking shall not read university, name, email or any self-declared attribute.	Metamorphic tests: changing these leaves outcome and score unchanged.	Must

4.4 Audit logging

Every automated rejection must be logged with a reason.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-L1 Log every decision	Every automated and human decision shall be appended to a decision log with outcome, tentative outcome, reason code, reason, score, criteria version, parser version, confidence, trace, actor and timestamp.	count(decision_log) ≥ count(applications); every rejection has reason_code.	Must
FR-L2 Append-only	The decision log shall be append-only: update and delete shall be refused by the storage layer.	SQL UPDATE/DELETE on decision_log raise an error.	Must
FR-L3 Export	Authorised users shall be able to filter the log (outcome, reason, job, actor) and export it as CSV.	CSV export contains every filtered row with 14 columns.	Should

4.5 Bias audit

Selection rate for any declared group must not fall below 80% of the highest group's rate.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-F1 Fairness rule (testable form)	For every declared group g with $n_g \geq 30$ decided applications in the audit window, $SR_g \geq 0.80 \times \max_h SR_h$, where $SR = \text{shortlisted} \div \text{decided}$. A group below this is a breach.	0.80 passes, 0.78 breaches; groups with $n < 30$ reported but not enforced.	Must
FR-F2 Recurring audit	The audit shall run monthly over a rolling 90-day window per job and company-wide, after every criteria change, and on demand.	One scheduled audit per scope per month in the audits table.	Must
FR-F3 Breach consequence	A statistically significant breach ($z \geq 1.96$) shall pause automated rejection for the job until a re-audit passes; a non-significant breach shall open a warning incident.	After a critical audit, new rejections for the job are routed to manual review.	Must
FR-F4 Self-identification	Declared groups shall come from voluntary self-identification stored separately from screening data, with "Prefer not to say" excluded from comparison.	self_id is read only by audit; 'Prefer not to say' has no ratio.	Must

4.6 Human review

Low-confidence parses go to a person.

ID	REQUIREMENT	ACCEPTANCE CRITERION (MEASURABLE)	PRIORITY
FR-R1 Review queue	Applications in manual review shall be listed oldest-first with the reason they were routed and the withheld automated outcome.	Queue shows reason code, confidence and tentative outcome per row.	Must
FR-R2 Verified re-screen	A reviewer shall correct the parsed fields; the same routine then decides with confidence 1, and the log records the reviewer as actor.	Reviewer decisions have actor 'reviewer:<name>'.	Must
FR-R3 AI-assisted extraction	A local small language model could propose field values; any value not found in the résumé text shall be discarded.	Hallucinated skills, cities and employers are dropped.	Could

4.7 Fairness requirement as an executable test

Given 100 decided applicants in group A with 50 shortlisted, and 100 in group B with 40 shortlisted, **when** the audit runs, **then** B's impact ratio is $0.40 / 0.50 = 0.80$ and the audit passes.

Given B instead has 39 shortlisted, **then** the ratio is 0.78, a **breach** is reported for B, and the job's severity is graded by significance (FR-F3).

Given B has only 10 decided applicants, **then** B is reported as low-sample and cannot trigger a breach.

5 Non-functional requirements

REQUIREMENT	SCOPE	TARGET	EVIDENCE
NFR-1 Performance	Screening decision for a parsed application	≤ 50 ms p95 (routine is $O(\text{skills})$); $OCR \leq 3$ s per page	Routine < 1 ms per call in unit tests; OCR 237 ms/page on an M2 laptop
NFR-2 Throughput	Volume from the case study	90,000 applications/year ≈ 360 per working day (250 days); peak $5 \times = 1,800/\text{day}$	Seed of 2,000 parses + screens in ≈ 4 min including OCR
NFR-3 Availability	Screening during migration	Screening never stops: legacy remains primary until each increment's exit criteria are met (D5)	Shadow mode runs both systems on the same PDFs

REQUIREMENT	SCOPE	TARGET	EVIDENCE
NFR-4 Auditability	Decision records	100% of decisions reproducible from (parse, criteria version, routine version)	Append-only log; versions on every row
NFR-5 Privacy	Personal & self-ID data	Self-ID visible to audit role only; never shown to recruiters; data minimised (DPDP Act 2023)	Separate table; not returned by application APIs
NFR-6 Maintainability	Rule change effort	A ranking rule change touches 0 files in parsing/; criteria change needs no deploy	Static import check; criteria are data
NFR-7 Usability	Recruiter task	Explain any decision in ≤ 2 clicks from the ranked list	Candidate page shows path, reason, rank keys
NFR-8 Portability	Deployment	Runs on a single host with Bun; SQLite file DB; no external services required	OCR and AI optional with graceful fallback

6 Use cases

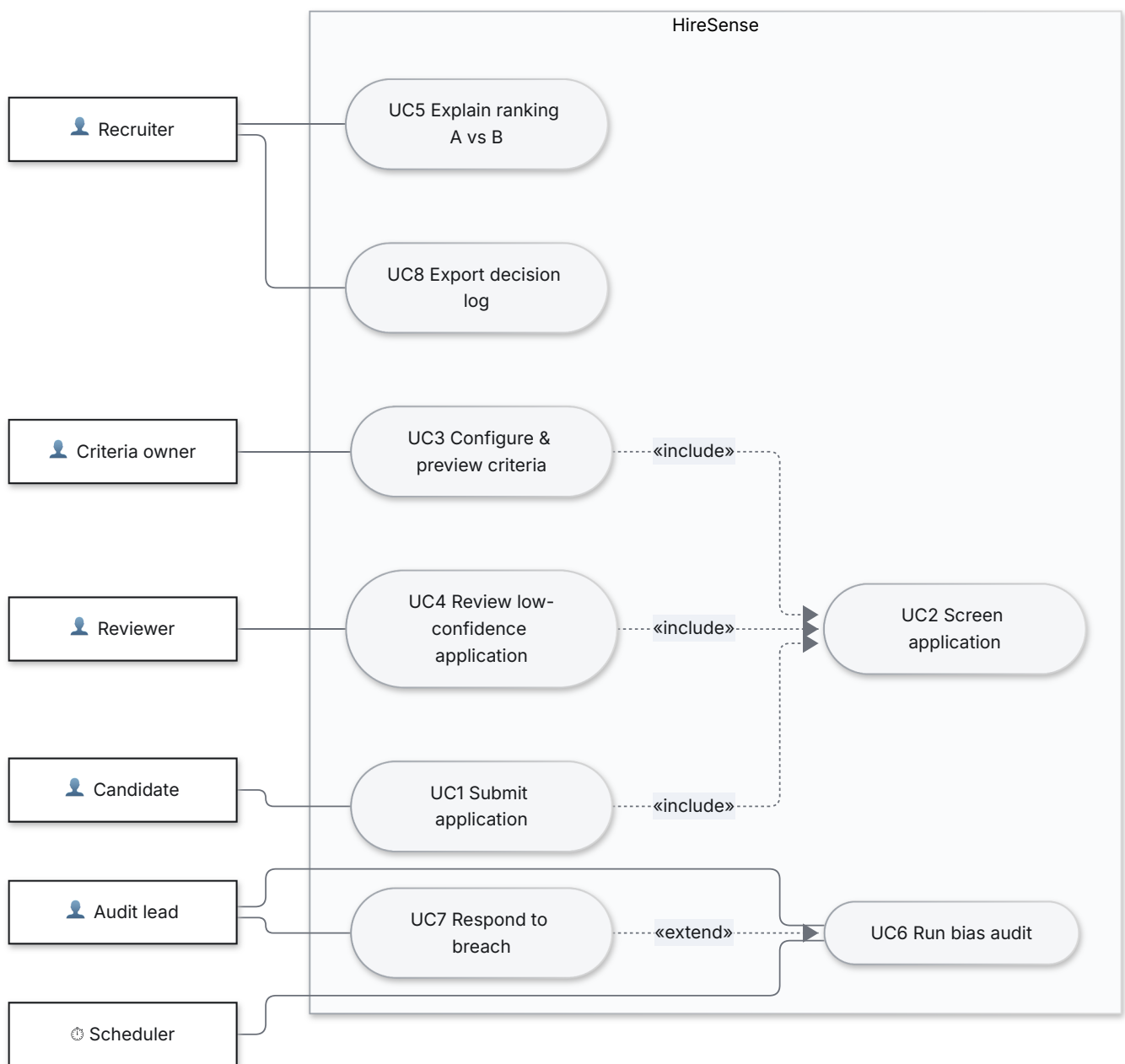


Figure 1. Use-case diagram. Screening (UC2) is included by submission, by review and by criteria publication. A breach response extends the audit.

UC	ACTOR	MAIN FLOW	ALTERNATIVE / EXCEPTION	REQS
UC1 Submit application	Candidate	Upload PDF + optional self-ID → stored → UC2	Not a PDF / > 5 MB → rejected with message	FR-P1, FR-F4
UC2 Screen application	System	Parse → (OCR if no text) → routine → policy → log → rank	Confidence < 0.6 → manual review; job paused → rejections to review	FR-P2, FR-S1, FR-L1
UC3 Configure criteria	Criteria owner	Edit → preview impact → publish with note → re-screen job → audit	Invalid bands/skills → error; no note → blocked	FR-C1–C3, FR-F2
UC4 Review application	Reviewer	Open queue → compare PDF and fields → (AI suggest) → confirm → routine decides	Model unavailable → manual entry	FR-R1–R3
UC5 Explain ranking	Recruiter	Pick A and B → first differing rank key + both paths	Different jobs → compare outcomes, not ranks	FR-S2, FR-S3
UC6 Run bias audit	Scheduler / audit lead	Monthly / on change / on demand → compute ratios → grade → record	Insufficient sample → reported only	FR-F1, FR-F2
UC7 Respond to breach	Audit lead	Warning → incident; critical → pause auto-reject → remediate → re-audit	Re-audit passes → resume	FR-F3
UC8 Export log	Recruiter / audit lead	Filter → CSV	—	FR-L3

7 Requirements traceability matrix

Each requirement traces forward to the design element that implements it and the automated test that verifies it (test references are `file:line` in `api/test/`). Requirements verified at system level are verified through the API and the review checklist.

REQ	SOURCE	DESIGN ELEMENT	VERIFIED BY	STATUS
FR-P1	CS: parsing quality	<code>parsing/parser.ts</code> · WEIGHTS	<code>parser.test.ts:20</code> <code>parser.test.ts:148</code>	Verified
FR-P2	CS: 29% fail silently	<code>parser.ts</code> · MIN_TEXT_CHARS, ANOMALY_CAP	<code>parser.test.ts:46</code> <code>parser.test.ts:56</code>	Verified
FR-P3	Work package WP-P	<code>parsing/ocr.ts</code> (PP-OCrv6 tiny)	<code>parser.test.ts:131</code>	Verified
FR-P4	CS: define acceptance threshold	<code>seed/evaluate.ts</code>	<code>parser.test.ts:68</code> <code>parser.test.ts:77</code>	Verified
FR-C1	CS: explicit and configurable	<code>types.ts</code> Criteria · <code>index.ts</code> validateCriteria	<code>routine.test.ts:108</code> <code>routine.test.ts:99</code>	Verified
FR-C2	CS: explainability	<code>criteria_versions</code> table	API smoke / review	System test
FR-C3	Preventive maintenance	<code>POST /jobs/:id/criteria/preview</code>	API smoke / review	System test
FR-C4	CS outcome: modular design	module boundaries (D3 §2)	API smoke / review	System test
FR-S1	CS: ranking routine	<code>ranking/routine.ts</code> <code>screen()</code>	<code>routine.test.ts:10</code> <code>routine.test.ts:18</code> <code>routine.test.ts:25</code> <code>routine.test.ts:32</code> <code>routine.test.ts:39</code>	Verified
FR-S2	CS: explainability	<code>Decision.trace</code> · <code>decision_log.trace</code>	<code>routine.test.ts:133</code>	Verified
FR-S3	CS: candidate 412 vs 87	<code>ranking/order.ts</code> <code>explainOrder()</code>	<code>audit.test.ts:81</code> <code>audit.test.ts:88</code>	Verified
FR-S4	CS: internal review (tier)	<code>self_id</code> table segregated	<code>routine.test.ts:121</code> <code>routine.test.ts:127</code>	Verified

REQ	SOURCE	DESIGN ELEMENT	VERIFIED BY	STATUS
FR-L1	CS: log every rejection	decision_log table · screening.ts	API smoke / review	System test
FR-L2	Auditability	SQLite triggers	audit.test.ts:101 audit.test.ts:102	Verified
FR-L3	Audit evidence	GET /decisions.csv	API smoke / review	System test
FR-F1	CS: audit requirement	audit/fairness.ts	audit.test.ts:16 audit.test.ts:22 audit.test.ts:29	Verified
FR-F2	CS: periodic bias audit	audit/service.ts runAudit	audit-service.test.ts:40	Verified
FR-F3	CS: defined consequence	service.ts severityOf · screening policy	audit-service.test.ts:48 audit-service.test.ts:61 audit-service.test.ts:69	Verified
FR-F4	Privacy	self_id table	'Prefer not to say' is never compared	Check
FR-R1	CS: manual review	GET /review	API smoke / review	System test
FR-R2	Accountability	POST /applications/:id/review	API smoke / review	System test
FR-R3	Reviewer productivity	parsing/llm.ts	llm.test.ts:47 llm.test.ts:40	Verified

Table 1. Coverage of the matrix

22 functional requirements · 15 verified by automated unit/integration tests · 7 verified by system tests through the API. Non-functional requirements are verified by measurement (§5).

8 Feasibility and validation

FEASIBILITY	ASSESSMENT
Technical	Feasible: a working prototype implements every Must requirement. Measured 99.0% parse success and 6 silent failures in 2,000 résumés; all automated tests pass.
Operational	Feasible: shadow mode and increments let recruiters keep working; the review queue load fell from 215 to 94 after OCR.
Economic	Open-source stack (Bun, Next.js, SQLite, PP-OCR, Qwen3 1.7B), no licence fees; runs on existing laptops/servers. Effort estimate in D5.
Legal / ethical	Self-ID is voluntary and segregated; adverse-impact monitoring follows the four-fifths rule; personal data minimised (DPDP Act 2023).

Requirement validation checklist

CHECK	RESULT
Every requirement has a unique ID and a measurable acceptance criterion	Yes
Case-study figures used verbatim (routine, 0.6, 29%, 80%)	Yes
Design parameters we chose are labelled as such ($n \geq 30$, $z \geq 1.96$, 90-day window, weights)	Yes
No requirement conflicts (e.g. 'never reject a low-confidence parse' vs routine): resolved by N9 overriding to manual review	Yes
Each Must requirement is traced to design and verification	Yes